A-A-53237
29 September 1987

COMMERCIAL ITEM DESCRIPTION

BATH, WATER, DENTAL IMPRESSION MATERIAL, Low Temperature

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers a low temperature, electrically heated, thermostatically controlled water bath suitable for tempering dental materials. The complete water bath and insert tray shall be furnished as specified below.

Class 1 - Complete water bath

Class 2 - Replacement insert tray

Salient Characteristics:

Materials. Materials shall conform to the applicable requirements specified herein. Materials not specifically designated shall be of high commercial quality throughout and shall be entirely suitable for the intended purpose.

Style, design and dimensions. Class 1 shall be a complete water bath and Class 2 shall be a replacement insert tray (basin) for the water bath. The style, design and dimensions of the water bath and the insert tray shall be in accordance with Defense Personnel Support Center (DPSC) Drawing No. 24376.

The water bath shall be a low temperature water bath and shall be capable of heating water to a thermostatically controlled temperature ranging from 90 to 160°F. The water bath shall be suitable for operation on 115 volt, 50/60 Hertz, alternating current. There shall be a wax warming device for attachment to the thermometer support. The device shall provide a moveable cup (raise, lower, or rotate to the side) for warming low fusing materials (wax).

Cabinet. The cabinet shall be a spot welded assembly of a frame, top, and a ground clip, together with the necessary angle and weld nuts. The frame and top shall be fabricated of No. 22 (.031 inch) U.S. Standard Gauge 18-8 stainless steel sheets. The exterior of the cabinet shall be finished in a No. 4 or brush finish.

AMSC/NA FSC 6520

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Heating element. The strip heater shall be attached in the well of the cabinet with screws and shall be spatially suspended by spacers. The strip heater shall be rated at 211 watts plus or minus 10 percent at 115 volt, 50/60 Hertz, alternating current.

The heater shall consist of nickel-chromium alloy wire, imbedded in compressed refactory material and enclosed in a chrome steel housing.

Main switch. The on-off switch shall be a four position, double pole, single throw, rotary switch with a 15 ampere, 125 volt alternating current rating and shall serve as a total disconnecting switch.

The on-off switch knob shall be keyed to the switch and imprinted "ON-OFF-ON-OFF" at the four reciprocating positions of the switch.

Thermostat. The thermostat shall be of the strut-type construction and shall be enclosed in a stainless steel case. The thermoswitch shall be mounted on the wall of the cabinet well and the adjusting stem shall protrude through an elongated hole in the front panel of the cabinet. The thermoswitch shall have a 10 ampere, 115 volt alternating current rating.

Dial. The dial may serve a multiple function as a temperature reference and as an identificaton nameplate. The dial shall be placed over the thermoswitch stem and shall be mounted on the front panel of the cabinet. The temperature numbers shall be printed at intervals of 20°F or less, starting at 100°F or less and ending at 160°F or more. The thermoswitch knob shall be attached to the thermoswitch stem and shall be rotatably adjustable to the temperature selections.

Pilot lights. The two pilot lamps shall be of the neon type with a nylon body and a red plastic lens. One pilot lamp shall indicate only when the thermoswitch contacts are closed and activating the strip heater. The other pilot lamp shall indicate continuously when the on-off switch is turned to "ON".

Bottom panel. The bottom panel shall be fabricated of No. 16 (.050 inch) B & S Gauge aluminum sheet. The bottom panel shall be securely attached to the cabinet. Each corner of the bottom panel shall have a securely attached leg with a polyethylene cap to properly contact the surface on which it is resting without wobble or other imbalance.

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Thermal protection. The thermal cut-off device shall be provided as a built-in safety factor that shall effectively provide a current interruption in the event of a thermoswitch malfunction. The device shall be mounted to the inside of the bottom panel below the strip heater. The device shall respond by opening the electrical circuit at a temperature of 200 - 220°F at its position inside the bottom panel.

The thermal cut-off shall interrupt the electrical circuit at 210 ± 10°F, when tested as specified herein.

Electrical cord. Each unit shall be provided with an electrical cord, no less than 6 feet in usable length. The cord shall be 3-wire, No. 16 or 18 gage, type SPT-3 thermoplastic insulated or type HPN, HSJ or HSJO rubber jacketed cord. A 3-prong vinyl plug shall be attached to one end and shall be suitable for use with receptables specified in American National Standard Institute (ANSI) Standard C73.11 for caps and receptacles; 125 volts, 15 amperes, 2 pole, 3-wire grounding type. The opposite end shall be slit back and stripped for attachment of terminals for connection to the on-off switch and to the ground clip. The water bath shall conform to the requirements of Underwriters Laboratories, Inc. standards.

Removable basin (insert tray). The removable insert tray, both when furnished as part of Class 1 and as a separate replacement part (Class 2), shall be formed from one piece of No. 22 (.031 inch) U.S. Standard Gauge AISI Type 304 stainless steel of FED-STD-66. The basin shall be furnished with a rolled bead on the top edge to provide a finger grip for removal. The basin shall contain a single embossed rib on its bottom for reinforcing and an embossed water level line on one side, 9/16 ± 1/16 inch from the top. The interior shall be satin finished and the exterior shall be finished by either electro-polish, bead blast, or any other method which shall impart a clean appearance not necessarily devoid of draw marks. The removable basin shall be placed into the well of the cabinet and shall rest on the strip heater and basin support attached to the bottom of the well.

The basin shall be lightly forced forward and into mechanical contact with the thermoswitch by two pair of slots at the rearward part of the well. When the material between the slots is deformed, the spring action shall exert and maintain sufficient force to hold the basin against the thermoswitch.

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Thermometer. The thermometer shall be 6 ± 1/2 inches long and 1/4 ± 1/32 inch in diameter. The thermometer shall be graduated from 0 to 220°F, engraved in 2° divisions and numbered every 20°. A rubber washer shall be placed over the thermometer to abut with the button top and a rubber tube shall be placed protectively over the mercury bulb. The thermometer shall be suspended in the loop of the thermometer holder which shall be rotatably adjustable in the attached 18-8 stainless steel holder support. The positioning of the thermometer in the holder shall allow for adequate immersion of the bulb in the properly filled basin.

Lifting platform. The lifting platform shall be fabricated of wire and shall be formed and twisted to accept an open end cloth bag cover. Two cloth covers shall be furnished with each water bath.

Performance.

Dielectric strength. The water bath shall be capable of withstanding, without failure as specified herein.

Temperature control. The thermoswitch knob shall be adjustable to compensate for differences between the dial and thermometer readings. The thermostat shall maintain the stabilized temperature of the water bath within plus or minus 3°F, when tested as specified herein.

Temperature of components and table top. The maximum temperature of the components and the table top shall be as specified in table I, when tested as specified herein.

TABLE I. Maximum temperature of components and table top.

Location	Maximum Temperature (Degrees Fahrenheit)
Table (below strip heater)	100
Bottom panel (center)	180
Front of Cabinet (center)	180
Top of Cabinet (at Basin)	180
Side of Cabinet (at wall)	180
Rear of Cabinet (at wall)	180
Thermoswitch Knob	138
Pilot Lamp Wires	200
Wires at Strip Heater	300
On-Off Switch	200
Cord Set (inside at Strain Relief	194

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Operational test. The water bath shall operate satisfactorily when tested as specified herein.

Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the Contracting Officer, two preliminary copies of his proposed technical manual for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the Contractor shall furnish 15 copies of the approved technical manual to the Contracting Officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of the contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the Contracting Officer. Waiver of submission of drafts and submission of 15 additional copies shall be at the option of the Government.

Identification marking. Each water bath shall have a corrosion-resisting metal or pressure sensitive tape nameplate permanently affixed thereto. The plate shall bear the model or style designation, voltage, watts, frequency (optional), and the name or registered trademark of the manufacturer. The markings on the plate shall be permanent and legible.

Workmanship. All exposed surfaces shall be smooth and free of sharp edges. Water baths shall be free from defects which detract from their appearance or impair their serviceability. Welds shall be clean and free from cracks and other imperfections. Screws shall be properly formed and tightened.

Unit. Each (EA). One bath water, dental impression material, as specified, constitutes one unit.

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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the sample sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection purposes shall be one water bath or insert tray as applicable. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, AQL 2.5, and sampling for tests (when applicable) shall be conducted at inspection level S-1, AQL 1.0.

Tests. Tests shall be conducted to determine compliance with specification requirements.

Chemical analysis. In the absence of certificates of quality for the raw material, chemical analysis shall be conducted in accordance with method 111 or method 112 of FED-STD-151. In case of dispute, method 111 shall be used for referee analysis.

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Thermal protection. On completion of the temperature of components and table top test, the thermoswitch shall be shunted out of the circuit. The unit shall continue to operate without thermostatic control and the water shall be permitted to evaporate completely. The unit shall be permitted to heat to the 200 to 220°F level measured at the thermal cut-off device. The temperature at which the electrical circuit is interrupted shall be observed and recorded.

Dielectric strength. The electrical components shall be tested after complete assembly and shall withstand, without failure, a dielectric test of twice the rated voltage plus 1,000 volts or a total of 1,250 volts applied for one minute. This test shall be made on each of the individual lines of the circuit, using the metal frame as the other leg or ground of the 1,250 volt test.

Temperature control. The basin shall be filled with water to the water level mark and the thermometer shall be placed in the holder to immerse the bulb of the thermometer in the water. The thermoswitch shall be adjusted progressively until 120°F is indicated on the thermometer and the water temperature has stabilized. The variation in temperature above and below the 120°F stabilized temperature observed on the thermometer shall be recorded.

Temperature of components and table top. A right angle simulator shall be constructed of solid hardboard such as 1/2 inch plywood. The table portion shall measure no less than 34 inches square. The rear and side walls shall be equal to the table in width and no less than 30 inches in height. The assembled room simulator shall be painted non-reflecting black. An iron-constantan thermo-couple shall be attached to metallic parts by brazing and to non-reflecting components by tape. The bath to be tested shall be placed in the corner of the room simulator at one inch from the vertical walls. The basin shall be filled with water to the water level line, the on-off switch shall be turned to "on", and the thermoswitch knob shall be rotated clockwise to the stop for maximum temperature condition. The voltage shall be regulated to 115-116 volts, 60 hertz alternating current, with an amperage of 1.6 to 2.0 and ambient temperature of 75 ± 3°F. The equilibrium temperatures shown by the thermocouple sensors at the locations enumerated in table I shall be observed and recorded after 90 minutes of operation.

Operational test. The water bath shall be given an operational test by turning the heater on with the thermostat set at 120°F and observing if all components, except thermal cut-off, are functioning properly.

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Drawings. The drawings forming a part of this specification are specification control drawings. The supplier shall be responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in improper design, the supplier shall provide a product with dimensions within the tolerances prescribed on the drawing to insure proper design, function, fit, assembly, and operation of the item. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

National Stock Number	Item Identification
6520-01-004-2658	BATH, WATER, DENTAL IMPRESSION MATERIAL, Thermostatically Controlled, 90° to 160°F, 115 volt, 60 Hz, AC
6520-01-222-0683	TRAY, INSERT DENTAL

Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-53237	BATH, WATER, DENTAL IMPRESSION MATERIAL, LOW TEMPERATURE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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